The Rule of Halves: the Holes of Every Close-Packed Stacking

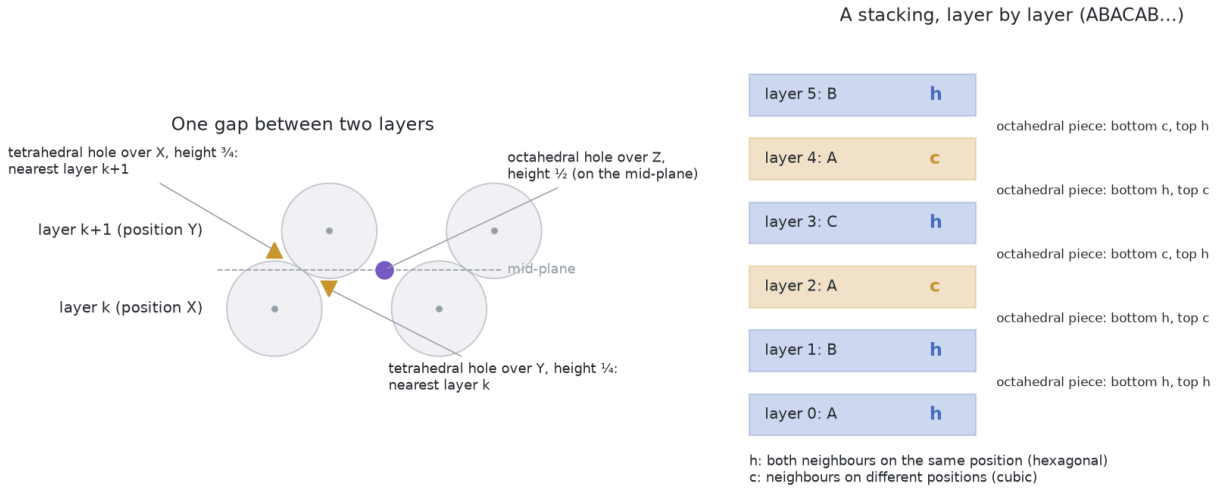


Theorem: in every close-packed stacking, along the whole weight walk, each tetrahedral hole's piece is the piece of the cubic or the hexagonal stacking named by its nearest layer, and each octahedral hole's piece is a top half and a bottom half, each the half of the cubic or the hexagonal piece named by the layer on that side; so the pieces of all stackings are finite in kind and an octahedral piece's volume is always the mean of its two parents' volumes.

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Paper #57 · Version 2 · 10 November 2026 (Version 1: 26 October 2026) · DICTO DOI: 10.5281/zenodo.23647898 · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/b24fd15d/studies/2026-10-25-stackings (research branch, commit b24fd15d); a snapshot is attached to this record
The ideas are DICTO's: following the web of bonds onto the other close-packed stackings (25 October 2026), and the names Chucky, Fang and Triskelion Tetrahedron, and the set name Janus c/h pieces. Computation, exact proofs and drafting with Claude (Anthropic); a first draft by a local open model was used for the order of sections only.

Version 2 corrects the notes on methods: the names of our computers and private file paths are removed, with small wording fixes. No result changes.

Abstract. Stack layers of balls and the gaps between them are small octahedra and tetrahedra. Stack them in the cubic way (ABCABC…) or the hexagonal way (ABAB…) and these gaps are arranged differently; crystals also use countless mixed stackings, such as ABAC or ABCACB. In papers #51, #53 and #56 we slid one number, a weight, between the two kinds of gap and followed how space is shared out among them: the web of bonds. Here we follow it on every stacking at once, and find a simple rule. A tetrahedral gap always takes the piece it would have in the pure stacking of the layer it sits closest to. An octahedral gap sits half-way between two layers, and its piece is two halves: the top half is the top half of the cubic or the hexagonal piece, as the layer above decides, and the bottom half likewise from the layer below. So every stacking, of any length or pattern, is built from the two pure ones, and the kinds of piece are finite: we list them all, with exact volumes. Where a cubic layer meets a hexagonal one, octahedral pieces appear that are half cubic and half hexagonal, which DICTO calls the **Janus c/h pieces**; two of them he named **Chucky** and **Fang**; and its volume is exactly the mean of its two parents'. The proof is exact, with one bound done by hand. The rule was not found in the sources checked; its classical counterpart for the balls themselves, that each ball's surroundings are set by its own layer, is the reason the h/c notation of crystallography exists.



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Figure 1. Left: one gap and its three holes. Right: a stacking layer by layer; a layer is h when both neighbours sit on the same position, c otherwise; each octahedral piece takes its bottom half from the layer below and its top half from the layer above.

1. Where it started

Paper #56 [1] followed the weight walk of #53 [2] on the hexagonal stacking and found DICTO's Sonsam and Lahdeli, Ian Dury and Blockhead, the hexagonal twins of the cubic pieces of #51 [3]. DICTO then proposed following the web of bonds onto the other close-packed stackings. On the double-hexagonal stacking ABAC and the six-layer ABCACB the walk mixed the cubic and hexagonal pieces and added two pieces neither had, which DICTO named Chucky and Fang; the nine-layer samarium stacking ABACACBCB added nothing new. That pointed to a local rule, which we prove here.

2. The set-up

> *Each gap gets a share of space; one weight decides how big a share the tetrahedral gaps get; slide it and the shares change shape.*

Unit balls sit in close-packed layers (triangular lattices of spacing 1) at heights kd, d = √(2/3); layer k sits on one of the three positions A, B, C, and no two consecutive layers share a position. Layer k is **h** if layers k − 1 and k + 1 sit on the same position and **c** otherwise (Jagodzinski's notation [4]). In the gap between layer k (position X) and layer k + 1 (position Y) there are three holes (Figure 1): an octahedral hole over the third position Z at height (k + ½)d, a tetrahedral hole over X at height (k + ¾)d and one over Y at height (k + ¼)d. The holes are the sites of a power diagram [5]: each site p, with weight w_p, owns the points x for which $|x - p|^2 - w_p$ is smallest; octahedral sites have weight 0 and tetrahedral sites weight λ. The pieces are convex and fill space; two pieces sharing a face are bonded. The walk is λ sliding over $-11/24 \le \lambda \le 3/8$, the range in which, in at least one of the two pure stackings, both kinds of piece are present (the hexagonal tetrahedral pieces shrink to points at −11/24, the cubic ones at −3/8, the octahedral pieces of both at 3/8). On each open stretch of λ between the finitely many **stops**, where some piece changes its faces, every corner of every piece moves on a straight line (Lemma 1). The pure stackings are the cubic ABC… (all layers c; the octet of #51) and the hexagonal AB… (all layers h; the twinned octet of #56).

Definitions. The *walk*: for each λ in the range we set $w_{tet} = λ$, $w_{oct} = 0$ and take the power diagram. A *piece*: the cell of one hole. A *stop*: a λ at which some piece gains or loses a face; a *stretch*: an open interval of λ between consecutive stops. A *window*: the six layers k − 2, …, k + 3 round the gap of layers k and k + 1, up to relabelling A, B, C; it is fixed by the h/c types of layers k − 1, …, k + 2, so there are 16. The *top* and *bottom halves* of an octahedral piece: its parts above and below the horizontal plane through its hole.

3. Main theorem

> **Theorem (the rule of halves).** For every close-packed stacking and every λ with $−11/24 ≤ λ ≤ 3/8$:
>
> 1. the piece of a tetrahedral hole is congruent to a tetrahedral piece of the pure stacking (cubic if c, hexagonal if h) of the type of its nearest layer: the hole over X, at height (k + ¾)d, is nearest to layer k + 1, the hole over Y to layer k;
> 2. the piece of an octahedral hole, cut by the horizontal plane through the hole, has a top half congruent to the top half of an octahedral piece of the pure stacking of the type of layer k + 1, and a bottom half congruent to the bottom half of an octahedral piece of the pure stacking of the type of layer k.
>
> **Corollaries.** (i) Up to congruence there are finitely many pieces in all close-packed stackings together: 21 combinatorial kinds, listed in the Appendix with exact volumes. (ii) The volume of an octahedral piece is the mean of the volumes of its cubic and hexagonal parents (or equal to the parent's, when both layers have the same type).

> *The two pure stackings are the only ingredients; every other stacking just decides, layer by layer, which ingredient each half comes from.*

4. Proof

Lemma 1 (corners move on straight lines). A corner of a piece is the common point of three dividing planes $2(q_i − p)·x = |q_i|^2 − |p|^2 − w_{q_i} + w_p$. Their normals do not depend on λ and their right-hand sides are affine in λ; on a stretch the same three planes define the corner, and their normals are linearly independent (otherwise they would not meet in a point), so the corner is $M^{−1}b(λ)$, with M the fixed matrix of the three normals and b(λ) affine: affine in λ. At a stop the set of planes through a corner may change, which is why the stretches are treated one by one. □

Step 1: six layers decide. The pieces of the holes of gap k depend only on the layers k − 2, …, k + 3. (i) *Exact computation.* Up to relabelling A, B, C, these six layers are fixed by the types of the four inner layers k − 1, …, k + 2: 16 windows. For each we built a periodic stacking containing the six layers and computed its power diagram exactly, in rational arithmetic, over the whole walk (the census engine of #53 [2], which also checks on every stretch that its finite list of sites suffices and that the pieces' volumes add up to the cell's). In every window, on every stretch, every corner of every piece of gap k lies within squared distance ½ of its hole (176 cases; squared distance along a straight path is convex in λ, so the stretch ends suffice). (ii) *By hand.* A hole three or more gaps away is at least 2.5d higher or lower than every hole of gap k, so $|q − p| ≥ 2.5d$, and $|w_p − w_q| ≤ 11/24$. Its dividing plane lies at distance $(|q − p|^2 + w_p − w_q)/(2|q − p|) ≥ ((2.5d)^2 − 11/24)/(5d) = (89/24)/(5d)$ from p (the function $t ↦ (t^2 − c)/(2t)$ increases), whose square is $(7921/576)/(50/3) = 23763/28800 >$

½. So for every λ no such plane meets the ball of squared radius ½ about p. (iii) *Convexity.* Let C be the piece of p computed from the holes of gaps k − 2 to k + 2 only. In any stacking with the same six layers the piece of p is C cut by the planes of farther holes; these miss the ball, and by (i) the piece in the test stacking lies inside the ball; a convex body inside the ball whose boundary there is made of C's planes only is C itself. So all such stackings give the same piece. This holds for every stacking, periodic or not, since any stacking shows one of the 16 windows round each of its gaps. □

Step 2: the 16 windows. For each window, stretch and hole of gap k we searched the 24 rigid motions that fix the hole's vertical axis and map the triangular layers to themselves (the 12 symmetries of the triangular lattice about the axis, with or without the reflection $z \mapsto -z$; they serve only to exhibit congruence). For tetrahedral holes, one such motion carries the corner functions (Lemma 1) of the piece exactly onto those of the stated pure piece: all cases. For octahedral holes, one motion carries the corner functions above, on and below the mid-plane, and the points where edges cross it (exact rational functions of λ), onto those of the stated pure halves: all 160 halves. Equal corner functions give equal pieces for every λ in the stretch. Together: 176 of 176 pieces, 0 failures; no stretch needed to be split where a corner crosses the mid-plane.

Step 3: stops. At a stop each piece is the limit of the pieces on the two adjacent stretches (the engine checks that the limit is the piece computed directly at the stop), so the congruences hold there too. □

The corollaries. (i) follows because the windows are finitely many and so are their stretches. (ii): each pure octahedral piece has a centre of symmetry at its hole (the octet's octahedral holes are centres of inversion, and so are the octahedral holes of hexagonal close packing), and inversion through the hole swaps the two halves, so each half has exactly half the parent's volume; by the theorem the top half of a mixed piece is congruent to the top half of one parent and its bottom half to the bottom half of the other, so its volume is half the one plus half the other. In particular, exactly: Fang = $5\sqrt{2}/324$, the mean of the cubic octahedron $\sqrt{2}/81$ and the hexagonal rhombohedron $\sqrt{2}/54$; Chucky = $383\sqrt{2}/768$, the mean of the cubic rhombic dodecahedron $\sqrt{2}/2$ and the hexagonal piece $191\sqrt{2}/384$.

Limits. The theorem covers $-11/24 \le \lambda \le 3/8$. At the ends of the range, and at $-3/8$ for the cubic tetrahedral pieces, some pieces shrink to points or edges; there the statement holds with those pieces read as points, as limits of the stretches. Steps 1(i) and 2 are exact computations; Step 1(ii) and (iii) and Lemma 1 are by hand; no floating-point arithmetic enters the proof.

Computational methods. Python with exact rationals (fractions, sympy), using the census engine of #53 [2]: for a periodic stacking it lists the holes within a fixed distance, computes each piece exactly at a rational λ, finds the stretch round it exactly, and expresses every corner as an affine function of λ, with checks that the site list suffices and that the volumes add up to the cell. All scripts and outputs are in the study folder [S]; a fresh copy of the repository on a second computer reproduced every result of Steps 1 and 2 and the exact volumes identically.

Use of AI. The computations, proofs and text were prepared with Claude (Anthropic) under DICTO's direction. An open language model, run locally, was used as an independent referee

of the proof and of this paper (it found no mathematical error and asked for clarifications now included) and drafted the order of sections; it wrote none of the final text.

5. The pieces

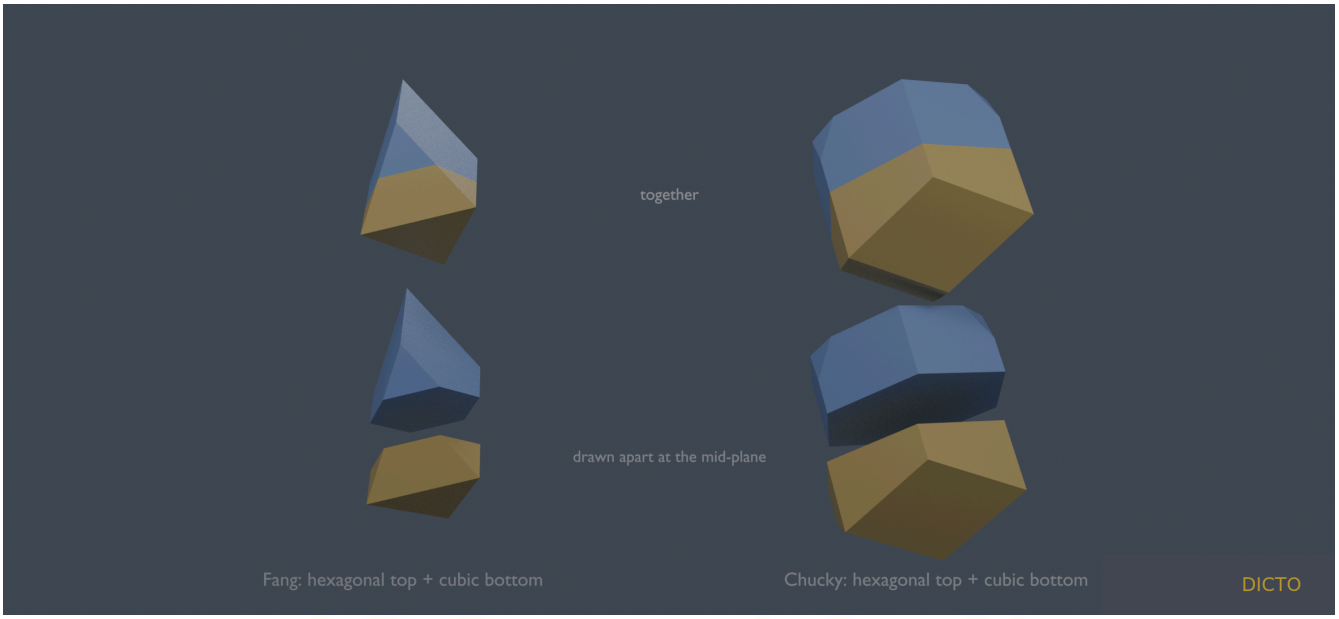


Figure 2. Fang (λ = 5/24) and Chucky (λ = −3/8) in the double-hexagonal stacking ABAC: hexagonal top half (blue), cubic bottom half (gold); whole above, drawn apart at the mid-plane below.

Chucky (DICTO's name): the octahedral piece at λ = −3/8 where a hexagonal layer lies above and a cubic layer below; 19 corners, with 3 triangles, 3 squares, 6 pentagons and one nonagon; volume 383√2/768. **Fang** (DICTO's name): the octahedral piece at λ = 5/24 in the same position; 7 corners, with 4 triangles and 3 squares; volume 5√2/324. Both belong to longer families of mixed pieces along their stretches (Appendix), and the stacking with the cubic layer above and the hexagonal one below gives their mirror images. DICTO calls all the octahedral pieces with one cubic and one hexagonal half the **Janus c/h pieces** (Janus, the two-faced god of doorways: each looks into one stacking above and the other below); Chucky and Fang are its named members, and two further families, of 9 and of 27 corners, complete it. **Triskelion Tetrahedron** (TT, DICTO's name): the tetrahedral piece of ABAC at λ = 5/24 with 16 corners (4 triangles, 6 hexagons). By the theorem it is the cubic piece there, a member of the sliding family of Samson in #51 [3]; the name marks that member; volume 79√2/324. The other pieces of the mixed stackings are the cubic and hexagonal pieces of #51 and #56; the two further Janus c/h families live on whole stretches and meet no stop.

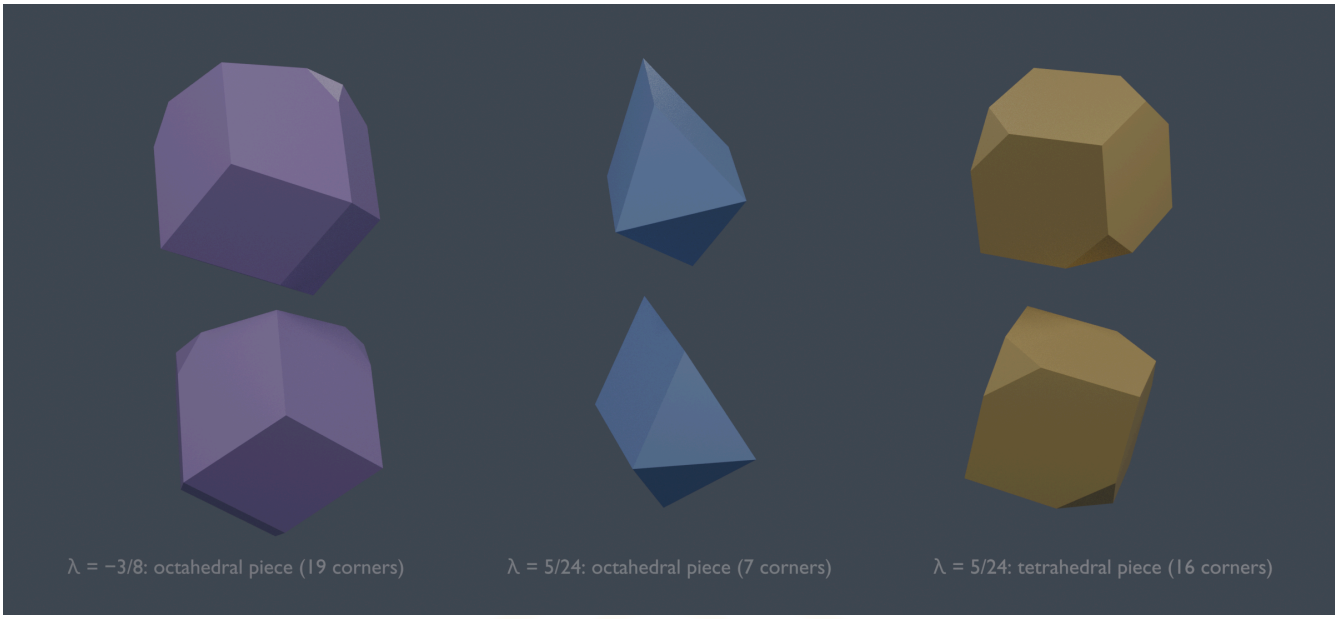


Figure 3. The three pieces of ABAC first met in this study, each from two sides: Chucky (19 corners), Fang (7 corners) and TT (16 corners, the cubic piece).

6. What is known, and what was not found

Known and credited. Close-packed stackings and their polytypes, and the h/c notation, whose purpose is that each ball's own surroundings are set by its layer's type [4]; our theorem is the counterpart for the holes, along the whole walk. Power diagrams [5]. The pure octet and twinned octet pieces [1, 3] and the walk [2]. Hales's proof of the Kepler conjecture divides the cubic and hexagonal packings into pieces built from their near-regular tetrahedra and octahedra ("decomposition stars"), for those two packings only [6]. Binary crystals of hard tetrahedra and octahedra: the cubic and hexagonal forms c-OT2 and h-OT2 (in h-OT2 octahedra sit on octahedra), which "can be combined in infinitely many different stacking variants", and small amounts of stacking faults observed in simulation [7]; that study has no cells round the holes and no rule relating mixed stackings to the pure forms. Radical (Laguerre) tessellations of binary hard-sphere crystals [8]. **Not found in the sources checked:** the pieces of the holes of mixed stackings, their finite catalogue, and the rule of halves. Searched: DICTO's reference library (eight queries on polytypes, hole cells, local environments, Delaunay and Voronoi cells of close packings, parameter families of power diagrams), five web searches, and Hales's three overview papers read for the purpose by a local model, which found no statement on mixed stackings or halves.

7. Open directions

- Other weights for the two kinds of tetrahedral hole (a three-kind walk, a map instead of a line), where the rule may split further.
- Keyed forms of Chucky and Fang, and a self-keyed Blockhead (#56).
- A world in the app to choose any stacking, slide λ and see every piece coloured by its parents.

References

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[S] DICTO, studies/2026-10-25-stackings (commit b24fd15d): stackings.py, catalogue.py, reach.py, exact_halves.py, exact_tets.py, exact_stretch.py, exact_proof2.py, exact_named.py, appendix.py; outputs exact_*.txt, appendix.json; proof-draft.txt, proof-additions.md, referee-120b.md, prior-art.md.

Appendix: every piece of every close-packed stacking

From appendix.json [S]. Columns: the kind of hole; the stretch of λ (open interval between two stops); what the piece is made from by the theorem (a cubic or hexagonal tetrahedral piece, or the types of an octahedral piece's top and bottom halves); the number of corners and the faces of the piece on that stretch; its exact volume as a function of λ, for balls of diameter 1. At a stop a piece is the limit of the adjacent stretches' pieces.

hole	stretch of λ	made from	corners	faces	volume
tetrahedral	(-11/24, -3/8)	hexagonal piece	4	4 triangles	$\sqrt{2}\cdot(24\cdot\lambda + 11)^3/6144$
tetrahedral	(-3/8, -1/8)	cubic piece	4	4 triangles	$\sqrt{2}\cdot(8\cdot\lambda + 3)^3/96$
tetrahedral	(-3/8, -1/8)	hexagonal piece	4	4 triangles	$\sqrt{2}\cdot(24\cdot\lambda + 11)^3/6144$
tetrahedral	(-1/8, 5/24)	cubic piece	16	4 triangles, 6 hexagons	$\sqrt{2}\cdot(512\cdot\lambda^3 - 576\cdot\lambda^2 + 216\cdot\lambda + 69)/384$
tetrahedral	(-1/8, 5/24)	hexagonal piece	19	6 triangles, 3 pentagons, 3 hexagons, 1 nonagon	$-\sqrt{2}\cdot(1536\cdot\lambda^3 + 5184\cdot\lambda^2 - 3384\cdot\lambda - 1013)/6144$
tetrahedral	(5/24, 3/8)	cubic piece	16	4 triangles, 6 hexagons	$\sqrt{2}\cdot(512\cdot\lambda^3 - 576\cdot\lambda^2 + 216\cdot\lambda + 69)/384$
tetrahedral	(5/24, 3/8)	hexagonal piece	19	9 squares, 4 hexagons	$\sqrt{2}\cdot(8\cdot\lambda + 1)\cdot(64\cdot\lambda^2 - 80\cdot\lambda + 37)/256$
octahedral	(-11/24, -3/8)	cubic top + cubic bottom	14	12 squares	$\sqrt{2}/2$
octahedral	(-11/24, -3/8)	cubic top + hexagonal bottom	19	3 triangles, 3 squares, 6 pentagons, 1 nonagon	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 1741)/6144$
octahedral	(-11/24, -3/8)	hexagonal top + cubic bottom	19	3 triangles, 3 squares, 6 pentagons, 1 nonagon	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 1741)/6144$
octahedral	(-11/24, -3/8)	hexagonal top + hexagonal bottom	24	6 triangles, 6 hexagons, 2 nonagons	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 205)/3072$
octahedral	(-3/8, -1/8)	cubic top + cubic bottom	30	8 triangles, 12 hexagons	$-\sqrt{2}\cdot(512\cdot\lambda^3 + 576\cdot\lambda^2 + 216\cdot\lambda + 3)/48$
octahedral	(-3/8, -1/8)	cubic top + hexagonal bottom	27	7 triangles, 9 hexagons, 1 nonagon	$-\sqrt{2}\cdot(46592\cdot\lambda^3 + 55872\cdot\lambda^2 + 22536\cdot\lambda - 13)/6144$
octahedral	(-3/8, -1/8)	hexagonal top + cubic bottom	27	7 triangles, 9 hexagons, 1 nonagon	$-\sqrt{2}\cdot(46592\cdot\lambda^3 + 55872\cdot\lambda^2 + 22536\cdot\lambda - 13)/6144$
octahedral	(-3/8, -1/8)	hexagonal top + hexagonal bottom	24	6 triangles, 6 hexagons, 2 nonagons	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 205)/3072$
octahedral	(-1/8, 5/24)	cubic top + cubic bottom	6	8 triangles	$-\sqrt{2}\cdot(8\cdot\lambda - 3)^3/192$
octahedral	(-1/8, 5/24)	cubic top + hexagonal bottom	9	5 triangles, 3 pentagons	$-\sqrt{2}\cdot(6656\cdot\lambda^3 - 14400\cdot\lambda^2 + 6840\cdot\lambda - 955)/6144$
octahedral	(-1/8, 5/24)	hexagonal top + cubic bottom	9	5 triangles, 3 pentagons	$-\sqrt{2}\cdot(6656\cdot\lambda^3 - 14400\cdot\lambda^2 + 6840\cdot\lambda - 955)/6144$
octahedral	(-1/8, 5/24)	hexagonal top + hexagonal bottom	12	2 triangles, 6 pentagons	$\sqrt{2}\cdot(1536\cdot\lambda^3 + 5184\cdot\lambda^2 - 3384\cdot\lambda + 523)/3072$
octahedral	(5/24, 3/8)	cubic top + cubic bottom	6	8 triangles	$-\sqrt{2}\cdot(8\cdot\lambda - 3)^3/192$
octahedral	(5/24, 3/8)	cubic top + hexagonal bottom	7	4 triangles, 3 squares	$-5\cdot\sqrt{2}\cdot(8\cdot\lambda - 3)^3/768$
octahedral	(5/24, 3/8)	hexagonal top + cubic bottom	7	4 triangles, 3 squares	$-5\cdot\sqrt{2}\cdot(8\cdot\lambda - 3)^3/768$
octahedral	(5/24, 3/8)	hexagonal top + hexagonal bottom	8	6 squares	$-\sqrt{2}\cdot(8\cdot\lambda - 3)^3/128$